Operating Systems Orange Level Problem Solving

NAME: G Purvi

SRN: PES1UG24CS570

Phase 1:

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

1A

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

1B

Phase 2:

```
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

2A

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5
c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

2B

Phase 3:

```
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "hello" > file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  test_objects.c
  untracked:  test_objects
  untracked:  Makefile
  untracked:  README.md
  untracked:  test_tree.c
  untracked:  commit.h
  untracked:  commit.c
  untracked:  tree.h
  untracked:  index.h
  untracked:  test_tree
  untracked:  file2.txt
  untracked:  pes.h
  untracked:  .gitignore
  untracked:  pes.c
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  tree.c

purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776430644 6 file1.txt
```

3A

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ cat .pes/index
100644 9df914ccc2d5c32d481e4ef31c17dcca996ec7df641b165f4d65a3d69802a9a7 1776431164 27 file1.txt
100644 2d3f6fef7953e2e36f82023a8ebe88d03bd974a0e4864913461b04fa6a17e9d0 1776431242 32 file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

3B

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "world" > file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  test_objects.c
  untracked:  test_objects
  untracked:  Makefile
  untracked:  README.md
  untracked:  test_tree.c
  untracked:  commit.h
  untracked:  commit.c
  untracked:  tree.h
  untracked:  index.h
  untracked:  test_tree
  untracked:  pes.h
  untracked:  .gitignore
  untracked:  pes.c
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  tree.c

purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

3A

Phase 4:

```
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "hello commit" > file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit "first commit"
error: commit requires a message (-m "message")
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit -m "first commit"
Committed: 2c52a3c8b33e... first commit
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes log
commit 2c52a3c8b33e7be765b4fc383003eb56ec75e35cbebd2b68da5bca3b37186f0a
Author: Purvi G <PES1UG24CS570>
Date:   1776430951

    first commit

purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

4A

```
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "hello commit" > file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit "first commit"
error: commit requires a message (-m "message")
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit -m "first commit"
Committed: 2c52a3c8b33e... first commit
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes log
commit 2c52a3c8b33e7be765b4fc383003eb56ec75e35cbebd2b68da5bca3b37186f0a
Author: Purvi G <PES1UG24CS570>
Date:   1776430951

    first commit

purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "line 2" >> file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit -m "update file1 - added line 2"
Committed: 3a3592610c1d... update file1 - added line 2
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "line 3" >> file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file1.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit -m "update file1 - added line 3"
Committed: 4378d442c33c... update file1 - added line 3
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "another file" >> file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit -m "added file2"
Committed: 972ddacaea82... added file2
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ echo "more changes" >> file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes add file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes commit -m "modified file2"
Committed: 980cddad3529... modified file2
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes log
commit 980cddad3529dd69321e28d2a790cc2ba4307b5e322a77ae5258de007a16cf3f
Author: Purvi G <PES1UG24CS570>
Date:   1776431255
```

4A

```
commit 980cddad3529dd69321e28d2a790cc2ba4307b5e322a77ae5258de007a16cf3f
Author: Purvi G <PES1UG24CS570>
Date:   1776431255

    modified file2

commit 972ddacaea82bdceb9a6a2c502e897b6ac638a0223d432343f3a056524223454
Author: Purvi G <PES1UG24CS570>
Date:   1776431227

    added file2le2

commit 4378d442c33cdb9fcc95c65722831e8387a5ed807595088d70ff48e25d1c68e6
Author: Purvi G <PES1UG24CS570>
Date:   1776431178

    update file1 - added line 3

commit 3a3592610c1de139bbb44f3e2e01d9dc0cc5e93a3ec761a468304dd940f5fb28
Author: Purvi G <PES1UG24CS570>
Date:   1776431136

    update file1 - added line 2

commit 2c52a3c8b33e7be765b4fc383003eb56ec75e35cbebd2b68da5bca3b37186f0a
Author: Purvi G <PES1UG24CS570>
Date:   1776430951

    first commitirst commit0> Q

purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ git add .
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ git commit -m "Phase 4: commit implementation with multiple commits"
[main ff7163f] Phase 4: commit implementation with multiple commits
 6 files changed, 197 insertions(+), 84 deletions(-)
 create mode 100644 file1.txt
 create mode 100644 file2.txt
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ git push
Username for 'https://github.com': purvigiridhar
Password for 'https://purvigiridhar@github.com':
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 2.44 KiB | 2.44 MiB/s, done.
Total 8 (delta 5), reused 0 (delta 0), pack reused 0
```

4A

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ ./pes log
commit 980cddad3529dd69321e28d2a790cc2ba4307b5e322a77ae5258de007a16cf3f
Author: Purvi G <PES1UG24CS570>
Date:   1776431255

    modified file2

commit 972ddacaea82bdceb9a6a2c502e897b6ac638a0223d432343f3a056524223454
Author: Purvi G <PES1UG24CS570>
Date:   1776431227

    added file2le2

commit 4378d442c33cdb9fcc95c65722831e8387a5ed807595088d70ff48e25d1c68e6
Author: Purvi G <PES1UG24CS570>
Date:   1776431178

    update file1 - added line 3

commit 3a3592610c1de139bbb44f3e2e01d9dc0cc5e93a3ec761a468304dd940f5fb28
Author: Purvi G <PES1UG24CS570>
Date:   1776431136

    update file1 - added line 2

commit 2c52a3c8b33e7be765b4fc383003eb56ec75e35cbebd2b68da5bca3b37186f0a
Author: Purvi G <PES1UG24CS570>
Date:   1776430951

    first commitirst commit0> Q
```

4A

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/1b/47d1a2ce9f15b39e5b4b385b185e4166f2bca2a54c5fb739d38c52190d784b
.pes/objects/27/0eceb8f52c98ab781645bbd8701376a7347dcd68afe6feb899a3abc46e15a3
.pes/objects/2c/52a3c8b33e7be765b4fc383003eb56ec75e35cbebd2b68da5bca3b37186f0a
.pes/objects/2d/3f6fef7953e2e36f82023a8ebe88d03bd974a0e4864913461b04fa6a17e9d0
.pes/objects/3a/3592610c1de139bbb44f3e2e01d9dc0cc5e93a3ec761a468304dd940f5fb28
.pes/objects/43/78d442c33cdb9fcc95c65722831e8387a5ed807595088d70ff48e25d1c68e6
.pes/objects/93/2a9248e0802e735d8107e5a4a7761ca789e9c9d3bd01a8472e4438b6dfeb42
.pes/objects/97/2ddacaea82bdceb9a6a2c502e897b6ac638a0223d432343f3a056524223454
.pes/objects/98/0cddad3529dd69321e28d2a790cc2ba4307b5e322a77ae5258de007a16cf3f
.pes/objects/9d/f914ccc2d5c32d481e4ef31c17dcca996ec7df641b165f4d65a3d69802a9a7
.pes/objects/c7/a347e8378e06755dda10fd32a38d2fda2b635d0c037b3812ffc7ef7e70b514
.pes/refs/heads/main
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

4B

```
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/1b/47d1a2ce9f15b39e5b4b385b185e4166f2bca2a54c5fb739d38c52190d784b
.pes/objects/27/0eceb8f52c98ab781645bbd8701376a7347dcd68afe6feb899a3abc46e15a3
.pes/objects/2c/52a3c8b33e7be765b4fc383003eb56ec75e35cbebd2b68da5bca3b37186f0a
.pes/objects/2d/3f6fef7953e2e36f82023a8ebe88d03bd974a0e4864913461b04fa6a17e9d0
.pes/objects/3a/3592610c1de139bbb44f3e2e01d9dc0cc5e93a3ec761a468304dd940f5fb28
.pes/objects/43/78d442c33cdb9fcc95c65722831e8387a5ed807595088d70ff48e25d1c68e6
.pes/objects/93/2a9248e0802e735d8107e5a4a7761ca789e9c9d3bd01a8472e4438b6dfeb42
.pes/objects/97/2ddacaea82bdceb9a6a2c502e897b6ac638a0223d432343f3a056524223454
.pes/objects/98/0cddad3529dd69321e28d2a790cc2ba4307b5e322a77ae5258de007a16cf3f
.pes/objects/9d/f914ccc2d5c32d481e4ef31c17dcca996ec7df641b165f4d65a3d69802a9a7
.pes/objects/c7/a347e8378e06755dda10fd32a38d2fda2b635d0c037b3812ffc7ef7e70b514
.pes/refs/heads/main
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ cat .pes/refs/heads/main
980cddad3529dd69321e28d2a790cc2ba4307b5e322a77ae5258de007a16cf3f
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

4C

Make test-integration:

```
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: be0cdacef78a... Initial commit

Log after first commit:
commit be0cdacef78a11ab956b747276e34f79c0af77114de381b2f5e23eb6e0dce94f
Author: Purvi G <PES1UG24CS570>
Date:   1776434057

    Initial commit


--- Second Commit ---
Committed: 4ed37eeaf01f... Update file.txt

--- Third Commit ---
Committed: 58ed298bc507... Add farewell

--- Full History ---
```

```
--- Third Commit ---
Committed: 58ed298bc507... Add farewell

--- Full History ---
commit 58ed298bc5073b12901e9bbfe94d7b28c0014fdc58788f8924482c181266f016
Author: Purvi G <PES1UG24CS570>
Date:   1776434057

    Add farewell

commit 4ed37eeaf01fe54923c9429e91bbd291d250d90380814a9687c0a8bdba3aa760
Author: Purvi G <PES1UG24CS570>
Date:   1776434057

    Update file.txt

commit be0cdacef78a11ab956b747276e34f79c0af77114de381b2f5e23eb6e0dce94f
Author: Purvi G <PES1UG24CS570>
Date:   1776434057

    Initial committial commitEQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
58ed298bc5073b12901e9bbfe94d7b28c0014fdc58788f8924482c181266f016

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/27/0eceb8f52c98ab781645bbd8701376a7347dcd68afe6feb899a3abc46e15a3
.pes/objects/4e/d37eeaf01fe54923c9429e91bbd291d250d90380814a9687c0a8bdba3aa760
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/58/ed298bc5073b12901e9bbfe94d7b28c0014fdc58788f8924482c181266f016
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/be/0cdacef78a11ab956b747276e34f79c0af77114de381b2f5e23eb6e0dce94f
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
purvi@purvi-virtual-machine:~/PES1UG24CS570-pes-vcs$
```

Phase 5 & 6: Analysis

Q5.1: A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout <branch> — what files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

A branch in PES (similar to Git) is a file in .pes/refs/heads/ containing a commit hash. To implement pes checkout <branch>, the system must update .pes/HEAD to point to the selected branch (i.e., ref: refs/heads/<branch>). Then, it must read the commit hash from that branch file, retrieve the corresponding tree object, and update the working directory by deleting existing files and recreating files from the tree.

This operation is complex because it must safely handle overwriting files, manage recursive directory structures, and ensure that uncommitted changes are not lost.

Q5.2: When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

To detect a dirty working directory, the system compares each file in the index with the current working directory using metadata such as file size and modification time (stat). If any differences are found, the file is considered modified.

During checkout, the system compares the current index with the target branch’s tree. If a file is modified locally and also differs in the target branch, a conflict occurs. In such cases, checkout must be refused to prevent data loss.

Q5.3: "Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

A detached HEAD state occurs when HEAD contains a commit hash directly instead of referencing a branch. In this state, new commits can still be created, but they are not associated with any branch.

As a result, these commits may become unreachable (dangling) if no branch points to them. To recover such commits, a user can create a new branch pointing to the commit hash. In real Git, the reflog can also help recover these commits.

Q6.1: Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

Garbage collection removes objects that are no longer reachable from any branch. The algorithm works by starting from all branch heads and traversing all reachable commits, trees, and blobs. Each visited object is marked as reachable.

A hash set can be used to efficiently track visited objects. After traversal, any unmarked objects are considered unreachable and can be safely deleted.

In a repository with 100,000 commits and 50 branches, the traversal may involve visiting hundreds of thousands of objects (including trees and blobs), typically around 200,000 to 500,000 objects.

Q6.2: Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this?

Running garbage collection concurrently with a commit operation is dangerous due to race conditions. For example, a commit process may create new objects but not yet update references (like HEAD or branch pointers). If garbage collection runs at this moment, it may not detect these new objects as reachable and delete them.

This leads to repository corruption, as the commit would reference missing objects. Git avoids this problem by using atomic operations (write-then-rename), locking mechanisms, and by only deleting objects that are sufficiently old, ensuring that in-progress operations are not affected.